

Fitness Challenge Community

Objectives

- Build a social fitness app where users log workouts, track progress, and compete in group challenges.
- Enable progress photo comparisons with side-by-side before-and-after views.
- Create leaderboards ranking participants by challenge points, streak days, or total activity.
- Integrate wearables APIs (Fitbit, Apple Health) to auto-sync steps and calorie data.

Deliverables

- A fully functional fitness community web and mobile application.
- Workout logging interface supporting custom exercises, sets, reps, and cardio metrics.
- Group challenge creator with join links, rules, duration, and leaderboard tracking.
- Progress photo gallery with side-by-side comparison and privacy controls.
- Wearables integration dashboard showing synced steps, calories, and heart rate data.

Requirements

- Frontend: React Native for mobile; React.js for web dashboard.
- Backend: Node.js/Express REST API with scheduled jobs for leaderboard recalculation.
- Wearables: Fitbit API and Apple HealthKit integration for automatic activity import.
- Media: Cloudinary for progress photo storage with client-side compression.
- Database: MongoDB for flexible workout and challenge schemas.

Advanced Features

- Wearables Sync: OAuth-based connection to Fitbit, Garmin, and Apple Health data streams.
- Group Challenges: Multi-participant challenges with join codes, scoring rules, and countdown timers.

- Leaderboards: Real-time ranked boards updating after each logged workout or synced activity.
- Progress Photos: Private gallery with timeline slider for visual transformation tracking.
- Workout Plans: Trainer-created plans assigned to users with progression tracking.
- Streak Tracking: Daily activity streaks with milestone badges and push notification reminders.
- Nutrition Logging: Calorie and macro tracking with a barcode scanner for packaged foods.
- Social Feed: Activity feed showing friends' completed workouts with likes and comments.
- Analytics Dashboard: Personal stats on volume, frequency, PRs, and body metrics over time.
- Mobile Responsiveness: Native app feel with offline workout logging and background sync.
- Payment Gateway: Charge for premium membership plans and trainer-assigned workout programs via Stripe.
- Challenge Entry Fees: Allow organizers to set paid entry fees for competitive challenges with prize pools.

Trainer Payouts: Automatically distribute subscription revenue to certified trainers via Stripe Connect.